CSE 4615:: Wireless Networks

Future Trends

Lecture 3

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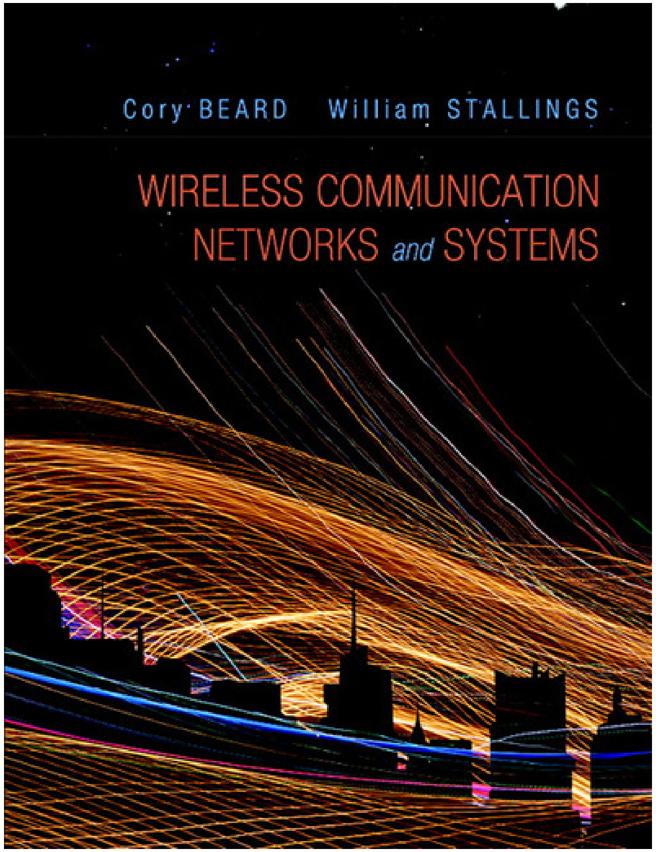
CHAPTER 1
INTRODUCTION



Wireless Communication Networks and Systems
1st edition
Cory Beard, William Stallings

Global cellular network

- Generations
  - 1G – Analog
  - 2G – Digital voice
    - Voice services with some moderate rate data services
  - 3G – Packet networks
    - Universal Mobile Phone Service (UMTS)
    - CDMA2000
  - 4G – New wireless approach (OFDM)
    - Higher spectral efficiency
    - 100 Mbps for high mobility users
    - 1 Gbps for low mobility access
    - Long Term Evolution (LTE) and LTE-Advanced

Mobile device revolution

- Originally just mobile phones
- Today's devices
  - Multi-megabit Internet access
  - Mobile apps
  - High megapixel digital cameras
  - Access to multiple types of wireless networks
    - Wi-Fi, Bluetooth, 3G, and 4G
  - Several on-board sensors
- Key to how many people interact with the world around them

Mobile device revolution

- Better use of spectrum
- Decreased costs
- Limited displays and input capabilities
- Tablets provide balance between smartphones and PCs
- Long distance
  - Cellular 3G and 4G
- Local areas
  - Wi-Fi
- Short distance
  - Bluetooth, ZigBee

Future trends

- LTE-Advanced and Gigabit Wi-Fi now being deployed.

Gigabit Wi-Fi :
- Ultra-fast wireless internet with speeds of **1 Gbps or more**
- Typically supported by **Wi-Fi 5 (802.11ac)** and **Wi-Fi 6 (802.11ax)**

Key Benefits:
- High-speed downloads and streaming
- Low latency for gaming and video calls
- Better performance in crowded networks
- Ideal for
  - **4K video and smart home** use (resolution of about **4000 pixels across the screen, sharper, clearer, and more detailed pictures — great for large screens)**,
  - **Online gaming**

Future trends

- Machine-to-machine communications
  - The "Internet of Things"
  - Devices interact with each other
    - Healthcare, disaster recovery, energy savings, security and surveillance, environmental awareness, education, manufacturing, and many others
  - Information dissemination
    - Data mining and decision support for Big Data
  - Automated adaptation and control
    - Home sensors collaborate with home appliances, HVAC systems, lighting systems, electric vehicle charging stations, and utility companies.
  - Eventually could interact in their own forms of social networking

Future trends

- Machine-to-machine communications
  - 100-fold increase in the number of devices
  - Type of communication would involve many short messages
  - Control applications will have real-time delay requirements
    - Much more stringent than for human interaction

Future trends

- Future networks
  - 1000-fold increase in data traffic by 2020
  - 5G – Started globally around **2019**
    - **High Speed**: Up to 10 Gbps
    - **Low Latency**: ~1 ms
    - **Massive IoT Connectivity**
    - **High Capacity in Crowded Areas**
    - **Network Slicing for Custom Services**
    - **Energy Efficient and Reliable**
  - **Ultra-low latency and massive IoT connectivity**
  - Foundation for smart systems, IoT, AR/VR, autonomous vehicles, and Industry 4.0.

Future trends:

Future networks

6G – Still in the research and development phase

Key Points:

- **Target Launch**: Around **2030** (globally expected).
- **Goal**: 10–100× faster than 5G (up to **1 Tbps** theoretical speeds).
- **Features**:
    - Ultra-low latency (~0.1 ms)
    - AI-integrated Semantic Communication
    - Real-time holograms & XR (Extended Reality)
    - Satellite-terrestrial integration
- **Research Countries**: USA, China, South Korea, Japan, EU nations are actively investing in 6G R&D.

Future trends

AI-integrated Semantic Communication

Future trends

- Future Technologies
  - Network densification – many small cells
  - Device-centric architectures - focus on what a device needs
  - Millimeter wave (mmWave) - frequencies in the 30 GHz to 300 GHz bands
    - Have much available bandwidth.
    - But require more transmit power and have higher attenuation due to obstructions
  - OFDM (Orthogonal Frequency Division Multiplexing
  - **MIMO (Multiple-Input Multiple-Output)**
  - Native support for machine to machine communication
    - Sustained low data rates, massive number of devices, and very low delays.

Political difficulties

- Between companies
  - Need common standards so products interoperate
  - Some areas have well agreed-upon standards
    - Wi-Fi, LTE
    - Not true for Internet of Things technologies
- Spectrum regulations
  - Governments dictate how spectrum is used
    - Many different types of uses and users
  - Some frequencies have somewhat restrictive bandwidths and power levels
    - Others have much more bandwidth available